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Editorial Office
Physical Review D
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Dear Editors,

I submit the manuscript "Dual Tetrad Gravity And The Field Origin: Deriving Fundamental Equations With $(3+1)^2$ Topology" for consideration as a Research Article in Physical Review D. The $(3+1)^2$ topology derived in the manuscript determines a covariant-complete, zero-parameter scalar-tensor Lagrangian, developed in full in the accompanying technical notes, whose predictions close bidirectionally across gravity, cosmology, and particle physics.

The question this paper addresses. Why does the gravitational constant take its measured value? No theory in the Standard Model or in General Relativity calculates G; it enters as empirical input. This paper shifts the focus from what the constants are to what structure requires them, bridging topology and fundamental physics. G follows from the coupling structure that GR itself requires for field closure.

Foundation in established physics. The derivation rests on four independent, well-cited results: the spinor double-cover requirement (Weinberg 1995, Geroch 1968), chiral decomposition of gravity (Ashtekar 1986), ghost-free bimetric gravity (Hassan–Rosen 2012), and cosmological stability analysis (Alexander, Marcianò, and Smolin 2014), with over 2,300 combined citations. All four arguments independently require two tetrahedra with opposite orientations. The stellated octahedron is the unique geometric realization satisfying all requirements simultaneously. Each integer in the resulting formula for G traces to a specific counting question about this structure: 4 from ADM constraints, 16 from the Hassan–Rosen interaction potential, and so on. No integers are chosen; they are counted from the geometry that established physics enforces.

Key results. From the $(3+1)^2$ dual tetrad topology, the paper derives:

1. The gravitational constant $G = \frac{17}{13}\,\alpha^{21}\,\frac{\hbar c}{m_e^2}$, predicted to 0.12% with zero adjustable parameters. The formula is sharply selective: changing the exponent by ± 1 shifts G by exactly $1/\alpha = 137$, so adjacent integers miss by over two orders of magnitude.

2. The dark matter correction factor $\mu(x) = x/(1+x)$, derived from detailed balance in field closure, with characteristic acceleration a_0 matching the empirical MOND scale to 0.41%. This resolves the flat rotation curve problem at galactic scales without invisible mass.

3. The electron mass m_e, derived from three overdetermined equations spanning gravity, cosmology, and electromagnetism, converging to 0.1% with no particle-physics input. The system also predicts $H = 69.92$ km/s/Mpc (matching the gravitational-wave standard siren measurement to 0.11%), and the round trip $m_e \to H \to m_e$ closes to 0.10%. Overdetermined systems generically have no solution; convergence indicates structural consistency.

4. The force hierarchy: gravity couples through 21 levels of the topology while electromagnetism couples through 1, giving $F_{\text{grav}}/F_{\text{EM}} \sim \alpha^{20} \approx 10^{-43}$.

Supplemental technical notes. Because these results touch several domains simultaneously, I have developed three self-contained companion technical notes to make each step of the argument

independently verifiable:

- *Covariant-Complete* $(3+1)^2$ *Gravity* presents the zero-parameter scalar-tensor action derived from the topology, its field equations (modified Poisson, modified Friedmann, and growth equations), and five zero-parameter predictions across gravity, cosmology, particle physics, galactic dynamics, and the CMB. The action is fully specified and falsifiable.
- *The Origin of Every Integer in G* traces each integer to its GR or bimetric gravity origin with a term-by-term translation table, demonstrating that 17, 13, and 21 are not fitted but counted from the stellated octahedron. It includes the sensitivity analysis showing that ± 1 in the exponent shifts G by exactly $1/\alpha$.
- *Why the Fine-Structure Constant Governs Gravitational Coupling* establishes why α appears in a gravitational formula (two independent routes, one electromagnetic and one gravitational, converge on the same characteristic scale r_e) and derives the constitutive law $\mu(x) = x/(1+x)$ from the coupling geometry.

A Python validation suite reproducing every numerical claim from CODATA 2022 constants is also included.

Scope and falsifiability. The paper acknowledges what the framework does not address: galaxy cluster dynamics, the full CMB power spectrum, the nuclear forces, and the cosmological constant. Every prediction is falsifiable through precision measurements, and the 0.12% residual in G (exceeding the $\sim$0.05% inter-experiment scatter) suggests either higher-order corrections or systematic effects.

Companion paper. This work extends results from "Deriving the Gravitational Constant from Spacetime Topology" (Nelson 2026), which derives G and electron spin from the stellated octahedron. The present paper expands the framework to nine additional equations, the dark matter resolution, and the electron mass derivation.

This work falls within PRD's scope of gravitation, general relativity, and modified theories of gravity. The connection to the Hassan–Rosen bimetric action in vielbein form, the covariant scalar-tensor completion, and the derived constitutive law place it in direct contact with active research programs in bimetric gravity and modified gravitational dynamics.

Thank you for your consideration.

Sincerely,

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